CPU hash enumeration: six-arm result

All 180 planned executions completed: 162 measured and 18 warmups, with byte-identical full certificates across arms for every fixture and repetition. The native analyzer accepted all 18 complete cells.

Job `hashed-ecdl-cpu-round18-20261030`, UID `68845702-fd9d-4940-b64c-67be6a25260c`, completed 2026-10-30 20:59:14 UTC. AMD EPYC 9R45, one software thread, 2 vCPU / 4 GiB allocation. L0 shared-host diagnostic; recorded times do not establish an isolated performance claim.

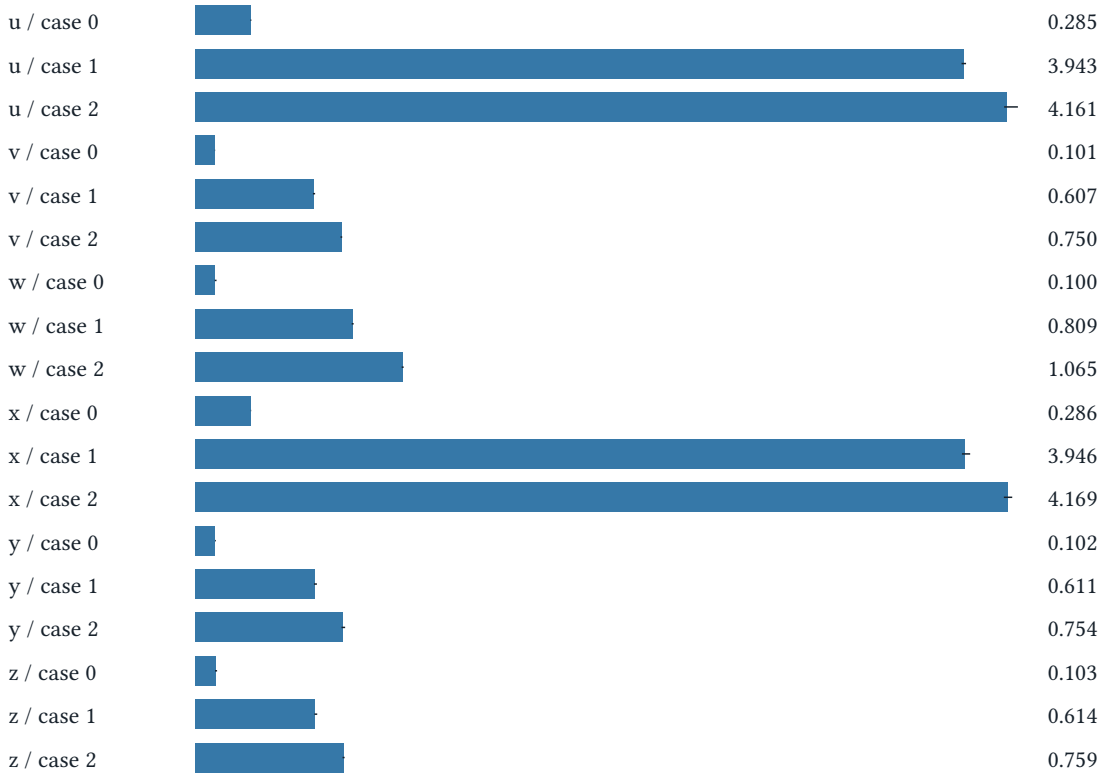
Arms u/x: original baseline `2bd4588fba60`; v/w: hash enumeration `0d6e3840edb6`; y: Curve context `23ff0e6bda63`; z: inverse cache `5d7dcb67616c`. Batch size 64 except w (1). Each cell has one warmup and nine measured repetitions. All comparisons below are within this panel.

Arm	Case	Min ms	Median ms	Max ms
u	0	0.283021	0.284861	0.287391
u	1	3.929944	3.942654	3.953055
u	2	4.148525	4.160665	4.218955
v	0	0.100051	0.101051	0.10173
v	1	0.603192	0.607232	0.614392
v	2	0.744722	0.750332	0.753313
w	0	0.099111	0.099961	0.108761
w	1	0.802062	0.808853	0.812043
w	2	1.058473	1.065143	1.068504
x	0	0.284721	0.285671	0.286531
x	1	3.932034	3.945584	3.974184
x	2	4.146975	4.168635	4.189905
y	0	0.10118	0.102471	0.105311
y	1	0.608482	0.611262	0.623702
y	2	0.749822	0.754313	0.768362
z	0	0.10125	0.102581	0.11167
z	1	0.610672	0.614282	0.625222
z	2	0.755213	0.759202	0.765742

Complete time includes curve and known-target construction, all trial setup and discarded trials, factor bases, query generation, reduction, internal verification, second independent replay and certificate serialization. Process launch, archive I/O and corrupted-scalar negative controls are outside this timer. The external process timer is retained separately in rows.csv.

Recorded complete latency

CPU complete recovery (ms), median and full range, n=9
Shared-host L0 diagnostics; no certified timing comparison



HASHED-ECDL-CPU-20261030 / raw rows.csv; labels decoded in report

Source: frozen `round18/analyze.rs`, recorded `round18/results/summary.csv`; nine measured values per cell. The graph shows medians, with the complete observed ranges in the preceding table.

Case 0: p=211, a=0, b=7, subgroup order 199, P=(3,33), m=3, tau=40, scalar=42, seed=111. Cases 1/2: p=65521, a=65518, b=3, subgroup order 65563, P=(0,64894); case 1 m=3, tau=80, scalar=45678, seed=222; case 2 m=4, tau=64, scalar=54321, seed=333. Limits: 16 trials and 1,000,000 work units. Known scalars construct fixtures; the solver receives public points.

Command: frozen `round18/run.sh`, executed with `EPHEMERAL_OUT=/work/results`. Baseline duplicate x provides the A/A control; all positive scalar checks, full replays and corrupted-scalar rejection completed. A/A ranges overlap in all three fixtures. No runs failed, timed out, resumed or were excluded.

Source, binary and returned-manifest identities are retained in `round18/results/`. Protocol SHA-256: `259ecd8eec7d7c17e99fa3f15c52196aa6b07cf33ac1bf0d60ab3c36f98e26a6`. Result archive SHA-256: `787fc9dd46c6a228d1622706e5834e7ad4f5cdec0a3b23afe396fa2c484bf13f`.

The cached-affine implementation in PR 2980 was implemented after the four revisions pinned in this panel and is not measured here. Its next measurement needs a separately frozen comparison. Reference material: the protocol's literature records and the retained round16 report informed this panel; no new literature or novelty claim is introduced.